SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A

sDry Run Evaluation

Code of Conduct:

I **Sadhana S (192524298)_** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SADHANA.S

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:

```
       A
      / \
     B   C
    /\   /\
   D  E F  G
```

Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue	Visited Node
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,

```
       A
      / \
     B   C
    /\   /\
   D  E F  G
```

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**

   1 3 6
   5 0 2
   4 7 8

Goal State :

1	**2**	**3**
4	**5**	**6**
7	**8**	**0**

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.
Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1			
2			
3			
4			
5			

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
Criteria	**Weightage (%)**	**Level 4 – Exemplary (4)**	**Level 3 – Proficient (3)**	**Level 2 – Developing (2)**	**Level 1 – Beginning (1)**
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

1. Breadth-First Search (BFS)

Graph

```
      A
    /   \
   B     C
  / \   / \
 D   E F   G
```

BFS Dry Run Table

Iteration	Queue (After Visiting Node)	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

Answer 1: BFS Traversal Order
A → B → C → D → E → F → G

Queue Contents After Each Iteration

Iteration	Queue
1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

2. Depth-First Search (DFS)

Graph

```
      A
    /   \
   B     C
  / \   / \
 D   E F   G
```

DFS Dry Run Table

Iteration	Stack	Visited Node
1	C, B	A
2	C, E, D	B
3	C, E	D
4	C	E
5	G, F	C
6	G	F
7	Empty	G

DFS Traversal Order
A → B → D → E → C → F → G

3. 8-Puzzle using Breadth-First Search (BFS)

Initial State
1 3 6
5 0 2
4 7 8
Goal State
1 2 3
4 5 6
7 8 0
Note: The given initial state is **not close to the goal** and requires many moves. Therefore, only the **first five BFS iterations** (as requested in the question) are shown.

BFS Dry Run

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	1 3 6 / 5 0 2 / 4 7 8	Initial State	4 new states generated
2	First generated state	Remaining 3 states	New child states added
3	Second generated state	Remaining states	New child states added
4	Third generated state	Remaining states	New child states added
5	Fourth generated state	Remaining states	New child states added

1. Which node is expanded in each iteration?

Iteration	Expanded Node
1	Initial State
2	First child of Initial State
3	Second child
4	Third child
5	Fourth child

2. How many states are generated in total?

- In the first expansion, the blank tile (0) is in the center.
- It can move **Up, Down, Left, Right**.
- Hence **4 child states** are generated initially.
- The total number of generated states continues to increase as BFS explores all possible states until the goal is found.

Answer: 4 states are generated in the first iteration (and additional states in later iterations until the goal is reached).

3. Write the complete solution path from the initial state to the goal state.
The solution path is the sequence of puzzle configurations from:
1 3 6

5 0 2
4 7 8
↓
(Intermediate states generated by BFS)
↓
1 2 3
4 5 6
7 8 0
Since this initial state requires multiple moves, the complete path contains several intermediate states. BFS guarantees that the first path found is the shortest one.

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?
Answer:
Breadth-First Search explores all states **level by level**. Every move in the 8-puzzle has the same cost (one move = one step). Therefore, BFS reaches the goal using the **minimum number of moves** before exploring deeper levels. Hence, BFS always returns the **shortest solution** in an unweighted puzzle.